YJC Annex B

Cover Page for College to submit with Prelim Examination Paper

Qn/No	Topic Set	Paper No.	Answers	Marks
1	Sequence	9740/01	(a) $L = \ln 2$	4
2	Binomial Expansion	9740/01	(i) $x^2 - 4x^3 + 12x^4 - 32x^5 + \ldots$; $\lvert x \rvert < \frac{1}{2},\ x \neq 0$ (ii) $(-2)^{r-2}(r-1)$	5
3	Complex Numbers	9740/01	(i) $\lvert w \rvert = 1$, $\arg(w) = \left(\frac{\pi}{6}\right), -1$ (ii) 0	6
4	Transformation of Curves	9740/01		5
5	Differentiation and its Application	9740/01	(a)(ii) $A = 2$ and $B = \frac{\pi}{2}$ (b) $\frac{20}{3}$	8
6	Differential Equations	9740/01	(i) $y = xe^{-x} - e^{-x} + 1$ (ii) $k = 1 - e^{-2}$	8
7	Tangent of Curve + Area under Curve	9740/01	(iii) $4\sqrt{a} - 12$	9
8	Functions	9740/01	(a) $x \geq 0$ or $x = -\sqrt{3}$ (b)(ii) 0 (b)(iii) $g^{-1} : x \to 2 - e^x$, $x \in [\ln 2, \infty)$	9
9	Vectors	9740/01	(b)(i) $\left(\frac{4}{7}, \frac{1}{3}, \frac{7}{3}\right)$ (b)(ii) $\sqrt{\frac{1}{3}}$ (c) $\mathbf{r} \bullet \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} = -1$	10
10	Curve Sketching	9740/01	(a) $x = -c$ and $y = ax + (-1 - ac)$ (b)(i) $b \geq 0$	9
11	AP/GP	9740/01	(a)(i) $a = 2m - 1,\ m \in Z^+$ That is, a may be any positive odd integer. (b)(i) \$41334.30 (b)(ii) \$200000 (b)(iii) 7	13
12	Integration and its Application	9740/01	(a)(i) $\frac{2}{3}(x-1)^{\frac{1}{2}}(x+2) + C$ (a)(ii) $\ln 3 - \frac{2}{9}$ (b)(ii) $\left(\pi - \frac{\pi^2}{4}\right)$	14

	Pure Maths			
1	Maclaurin's Series	9740/02	$-\frac{1}{2}x^2-\frac{1}{12}x^4-\frac{1}{45}x^6+\ldots$; $\ln 2 \approx \frac{\pi^2}{18}+\frac{\pi^4}{972}+\frac{\pi^6}{32805}$	5
2	System of Linear Equations	9740/02	(ii) $50 \le k \le 60$ (iii) $590000	7
3	Complex Numbers	9740/02	(a) Circle center (–4, 1) with radius $2\sqrt{3}$. (b)(ii) $\arg(z-\mathrm{i}) = -\frac{2\pi}{3}$; $z=-1+\mathrm{i}\left(1-\sqrt{3}\right)$; $m=\sqrt{3}$	8
4	Vectors	9740/02	(a)(i) $\begin{pmatrix}0\\1\\3\end{pmatrix}$ or equivalent (a)(ii) $y=\frac{z-2}{3}, x=0$ (b)(i) $b=3$, $a=-9$ (b)(ii) $a^2+b^2=314$	9
5	Method of Differences + Mathematical Induction	9740/02	(b) $\frac{3}{4}\left(1+(2n-1)3^n\right)$	11
	Statistics			
6	Binomial and Poisson Distributions	9740/02	(i) 0.371 (ii) 0.513 (iii) 0.933 (iv) 8	10
7	Normal Approximation to Binomial Distribution	9740/02	(b)(i) 0.356 (b)(ii) 0.842	10
8	Permutations and Combinations	9740/02	(a)(i) 40320 (a)(ii) 576 (b)(i) 25401600 (b)(ii) 9979200	8
9	Sampling and Hypothesis Testing	9740/02	(i) 8.76 , 0.860 (ii) p-value 0. 03359378 < 0.05, we reject H_0 … Probability of concluding that the mean studying time is less than 9 hours when it is actually 9 hours is 0.05. $\mu_0 < 8.50$ or $\mu_0 > 9.02$	12
10	Probability	9740/02	(i) …not mutually exclusive (ii) …not independent (iv) $p=\frac{2}{3}$	8
11	Linear Regression / Correlation	9740/02	(a) $(\bar{x},\bar{y})$ (b)(ii) x = 10.1. The predicted value of x is likely to be reliable as estimation is done by interpolation and r = 0.985 is close to 1.	12